Auteur: Siebe Haché
Studierichting: Informatica
Oefeningenreeks 4A nr. 60.5

$$\int \frac{\sqrt{x}}{\sqrt[4]{x}-1}dx \overset{x=t^4}{=} \int \frac{4t^5}{t-1}dt \overset{u=t-1}{=} 4\int \frac{(u+1)^5}{u}du$$
$$= 4\int \frac{u^5+5u^4+10u^3+10u^2+5u+1}{u}du$$
$$= 4\int \left(u^4+5u^3+10u^2+10u+5+\frac{1}{u}\right)du$$
$$= 4\left(\int u^4du + \int 5u^3du + \int 10u^2du + \int 10udu + \int 5du + \int \frac{1}{u}du\right)$$
$$= \frac{4u^5}{5} + \frac{20u^4}{4} + \frac{40u^3}{3} + \frac{40u^2}{2} + 20u + 4\ln|u| + c$$
$$\overset{u=t-1}{=} \frac{4\left(t-1\right)^5}{5} + \frac{20\left(t-1\right)^4}{4} + \frac{40\left(t-1\right)^3}{3} + \frac{40\left(t-1\right)^2}{2} + 20\left(t-1\right) +$$
$$4\ln|(t-1)| + c$$
$$\overset{t=\sqrt[4]{x}}{=} \frac{4\left(\sqrt[4]{x}-1\right)^5}{5} + \frac{20\left(\sqrt[4]{x}-1\right)^4}{4} + \frac{40\left(\sqrt[4]{x}-1\right)^3}{3} + \frac{40\left(\sqrt[4]{x}-1\right)^2}{2} +$$
$$20\left(\sqrt[4]{x}-1\right) + 4\ln\left|\left(\sqrt[4]{x}-1\right)\right| + c$$

References